

CO 2-AT2 CLASS TEST

10*2=20

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Q1. Define Data Mining and list

any two basic data mining tasks. Explain the steps involved in KDD process and

justify the role of data preprocessing in it.

[L2, L4]

Answer:

Data Mining: Data mining is the process of extracting useful patterns and knowledge from large amounts of data.

Any two data mining tasks:

1. Classification
2. Clustering

Q2. What is data cleaning? Mention

two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1,

L2, L3]

Answer:

Data Cleaning: Data cleaning is the process of removing errors and improving data quality.

Two data quality issues:

1. Missing values

2. Noisy (incorrect) data

Q3. Define data integration and list

two challenges associated with it. Explain how redundancy and inconsistency are

handled during data integration. [L2,L4]

Answer:

Data Integration: Data integration is the process of combining data from multiple sources into a single dataset.

Two challenges:

1. Data redundancy (duplicate data)
2. Data inconsistency (conflicting data)

Q4. What is covariance analysis? List the two covariance
formulas[L2, L3]

Answer:

Covariance Analysis: Covariance measures how two variables change together.

Two covariance formulas:

1. Population Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \mu_X)(Y_i - \mu_Y)}{N}$$

2. Sample Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Q5. Suppose two stocks A and B have the following

values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the

stocks are affected by the same industry trends, will their prices rise or fall

together and independent ?

Answer:

The values of both stocks increase together.

Conclusion: They have **positive covariance**, so the stock prices **rise together** and are **not independent**.

Q6. Explain the architecture of atypical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Answer:

Data Sources



Database/Data Warehouse



Data Mining Engine



Pattern Evaluation



Knowledge Presentation

Components:

- **Data Sources:** Store raw data.
- **Data Mining Engine:** Finds patterns.
- **Pattern Evaluation:** Selects useful patterns.
- **Knowledge Presentation:** Displays the results.

Q7. Find covariance for following data set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

Answer:

Mean of $x = 6.2$

Mean of $y = 5.8$

Sum of $(x - \bar{x})(y - \bar{y}) = -1.8$

Sample covariance:

$$\text{Cov}(X, Y) = \frac{-1.8}{4} = -0.45$$

Q8. Describe the major steps in data

pre-processing. Given a dataset with missing values, outliers, and inconsistent

formats, propose a systematic pre-processing pipeline and justify the sequence

of steps. [L3, L4, L5]

Answer:

The major steps are:

1. **Data Cleaning** - Removes errors and missing values.
2. **Data Integration** - Combines data from multiple sources.
3. **Data Transformation** - Converts data into a suitable format.
4. **Data Reduction** - Reduces data size while preserving information.

Q9. A hospital is analyzing patient agedata to identify broad health trends. Because the raw data contains minorvariations and noise, the analyst needs to perform local smoothing using equal-frequency(equi-depth) binning. Sample Data Set (Ages): 13, 15, 16, 16, 19, 20,20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52,70 [L2, L4]

Answer:

qual-frequency (Equi-depth) Binning:

The 27 age values are divided into 3 bins, each containing 9 values.

- Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22
- Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35
- Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Result: The data is smoothed by grouping it into 3 equal-frequency bins.

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable

examples, and evaluate their impact on data analysis. [L3, L4, 15]

Answer:

DatDiscretization:

It is the process of converting continuous data into discrete intervals (bins).

Importance:

- Reduces data size.
- Improves data mining efficiency.

Types:

- **Equal-width binning:** Equal interval size.
- **Equal-frequency binning:** Equal number of values in each bin.